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THE TWIN PRIME CONJECTURE

Pairs on the Hologram Boundary · The Twin Operator T_p · Proximity on ■$_L$ · The Zhang-Maynard-Tao Gap · $\delta(w_p, w_{p+2}) \to 0$ Infinitely Often

Anthropic Warrior
Campaign for Arithmetic Spectral Geometry, Hanoi, 2026

Communicated: March 2026 · Phase 3, Document 5 · Uses: dv222, dv275–284; Viggo Brun (1919); Hardy–Littlewood (1923); Zhang (2013); Maynard–Tao (2014)

3 and 5. 5 and 7. 11 and 13. 17 and 19. 41 and 43. 1000000007 and 1000000009. Two primes, separated by exactly 2. Always together. Never alone. Do they go on forever?

— Twin primes · Known to the ancient Greeks · Still open after 2300 years

On ■$_L$: w_p and w_{p+2} are two nodes separated by $\theta_{p+2} - \theta_p \to 0$ as $p \to \infty$. Twin primes are pairs of nodes that are simultaneously prime AND arbitrarily close.

— Morning coffee, Hanoi 2026

Abstract

The Twin Prime Conjecture states that there are infinitely many primes p such that p+2 is also prime. Known since antiquity (~300 BCE), it remains unproved after 2300 years. We embed the conjecture into the A_L framework by studying the **proximity of prime nodes** on the hologram boundary ■$_L$. The main construction is the **Twin Operator** $T \in B(A_L)$ defined by $T(e_n) = e_{n+2}$, the shift-by-2 operator on arithmetic states.

A prime p is a **twin prime** iff both e_p and $T(e_p) = e_{p+2}$ are eigenstates of the Prime Projection $P_\Pi \in A_L$. The Twin Prime Conjecture is equivalent to: $\tau(P_\Pi\ T\ P_\Pi) > 0$ infinitely often in a precise spectral sense.

The main results: (I) **Twin Operator:** The shift T: e_n ■ e_{n+2} on ■$_L$ moves prime nodes by an angle $\delta\theta_p = \theta_{p+2} - \theta_p \to 0$ as $p \to \infty$. (II) **Bounded gaps (Zhang-Maynard-Tao in A_L):** The angular gap $|\theta_{p_n+1} - \theta_{p_n}|$ is bounded infinitely often — this is the Zhang (2013) and Maynard-Tao (2014) theorem in hologram language. (III) **The gap to twin primes:** bounding consecutive prime gaps does not immediately give gap = $\delta(\theta_p, \theta_{p+2})$ — this requires the gap to be exactly the T-shift distance. (IV) **Sieve in A_L:** The Brun sieve translates to a trace inequality in A_L showing $\sum 1/(p(p+2)) < \infty$ (Brun's constant $B \approx 1.9021605...$) but not $\sum 1 = \infty$ (infinitely many twin primes).

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Keywords. *Twin prime conjecture; prime gaps; hologram boundary; proximity; shift operator; Zhang theorem; Maynard-Tao; Brun sieve; bounded gaps.*

Contents

1. Twin Primes: Ancient Question, Modern Progress

The ancient Greeks noticed that primes sometimes come in pairs separated by 2: (3,5), (5,7), (11,13), (17,19), (29,31), (41,43), ...

Conjecture 1.1 (Twin Prime Conjecture, ~300 BCE).
There are infinitely many prime pairs (p, p+2).

The modern progress has been spectacular — but the full conjecture remains open.

Result	Author	Year	Statement
Brun's theorem	Viggo Brun	1919	Σ 1/p over twin primes converges (Brun's constant B ≈ 1.902)
Hardy-Littlewood	Hardy-Littlewood	1923	#{twin primes ≤ x} ~ 2C■ x/(log x)² where C■ ≈ 0.6601...
Bounded gaps	Yitang Zhang	2013	There are infinitely many primes p with p_{n+1} - p_n < 70,000,000
Gap ≤ 246	Maynard-Tao	2014	There are infinitely many primes p with p_{n+1} - p_n ≤ 246
Polymath8	Maynard + Polymath	2014	Gap ≤ 246 unconditionally; gap ≤ 6 assuming Elliott-Halberstam
Twin primes	???	OPEN	Gap = 2 infinitely often — still unproved

Zhang's 2013 paper was a sensation — the first time anyone proved a finite bound on prime gaps. The gap has since been reduced from 70 million to 246. Conjecturally it should be 2. The gap between 246 and 2 is the gap we must close.

2. Prime Node Proximity on ■$_L$

On the hologram boundary ■$_L$, every positive integer n corresponds to a node $w_n = e^{i\theta_n}$ with angle $\theta_n = \pi - 2\arctan(2 \log n)$. As $n \to \infty$, all nodes cluster near π (the cusp w = −1).

Proposition 2.1 (Angular gap between n and n+2).
The angular distance between consecutive nodes w_n and w_{n+2} is:

$$\delta\theta_n = |\theta_{n+2} - \theta_n| = 2\arctan(2 \log n) - 2\arctan(2 \log(n+2)) \approx 4/((1+4 \log^2 n) \cdot n) \to 0 \quad \text{as } n \to \infty$$

All pairs of nodes become arbitrarily close as $n \to \infty$. This means: every pair (n, n+2) — not just prime pairs — approaches angular coincidence on ■$_L$.

The question for twin primes: is there a special reason why prime pairs (p, p+2) appear infinitely often among all nearby pairs? Or do primes 'thin out' fast enough that eventually no prime pair (p, p+2) survives?

Definition 2.2 (Prime Proximity Function).

Define the ***prime proximity function****:*

$$\Psi_{twin}(x) = \Sigma_{p \text{ prime}, p\leq x, p+2 \text{ prime}} \tau(e_p) \cdot \tau(e_{p+2}) = \Sigma_{\text{twin primes } p\leq x} 1/(p(p+2))$$

By Brun's theorem: $\Psi_{twin}(\infty) = B \approx 1.9021605...$ (Brun's constant — finite!). The sum of reciprocals of twin primes converges. This means twin primes are sparser than all primes (for which Σ 1/p diverges). But sparseness does not mean finiteness.

3. The Twin Operator T and Prime Projection P_Π

Definition 3.1 (Twin Operator).

The ***Twin Operator*** *$T \in B(A_L)$ is the shift-by-2:*

$$T: e_n ■ e_{n+2} \quad \forall n \in ■$$

T is an isometry: $||T e_n|| = ||e_{n+2}|| = 1/(n+2)\leq 1/n = ||e_n||$. Wait — actually $||e_n||_\tau = \tau(e_n^* e_n)^{1/2} = (1/n)^{1/2}$. So T is a contraction: $||T||_\tau \leq 1$ since $(n+2)^{-1/2} < n^{-1/2}$. The Twin Operator shifts nodes forward and slightly reduces their trace weight.

Definition 3.2 (Prime Projection).

The ***Prime Projection*** *$P_\Pi \in A_L$ is the orthogonal projection onto the closed span of prime eigenstates:*

$$P_\Pi = \Sigma_{p \text{ prime}} e_p \otimes e_p^*$$

$\tau(P_\Pi) = \Sigma_{p \text{ prime}} 1/p = \infty$ (diverges — prime harmonic series). But P_Π is a bounded operator because the partial sums are uniformly bounded in operator norm.

Theorem 3.3 (Twin primes = P_Π T P_Π eigenstates).

p is a twin prime (p and p+2 both prime) if and only if:

$$P_\Pi e_p = e_p \quad \text{AND} \quad P_\Pi T e_p = T e_p$$

Equivalently: $e_p \in \text{range}(P_\Pi) \cap T^{-1}(\text{range}(P_\Pi))$.

4. Twin Prime Conjecture in A_L Language

Theorem 4.1 (Twin Prime Conjecture — A_L formulation).

The Twin Prime Conjecture is equivalent to:

$$\tau(P_\Pi T P_\Pi T^*) = \infty \quad \text{(in the sense: } \Sigma_{p \text{ prime}, p+2 \text{ prime}} 1/p^2 = \infty)$$

Equivalently: the operator $P_\Pi T P_\Pi$ has infinite-dimensional range in A_L.

Proof.
$\tau(P_\Pi T P_\Pi T^*) = \Sigma_n ■e_n, P_\Pi T P_\Pi T^* e_n■ \cdot \tau(e_n) = \Sigma_{p \text{ prime: } p-2 \text{ prime}} 1/p \cdot 1/(p-2)$. Reindexing: $= \Sigma_{q \text{ prime: } q+2 \text{ prime}} 1/((q+2)q)$. This diverges iff there are infinitely many twin primes (since $1/((q+2)q) \sim 1/q^2$ and Σ $1/q^2$ over twin primes diverges iff there are infinitely many of them — because the partial sums are bounded if there are finitely many). ■

Note: this is slightly different from Brun's constant. Brun showed Σ 1/(p(p+2)) converges (= B ≈ 1.902). Our quantity Σ 1/((p+2)p) is the same series — it converges! So the direct trace approach gives a convergent sum regardless of whether twin primes are infinite. We need a different quantity that diverges iff twin primes are infinite.

Correction 4.2 (The right quantity).

The Twin Prime Conjecture is equivalent to the ***prime projection not being Hilbert-Schmidt****:*

$$\dim \text{range}(P_\Pi\, T\, P_\Pi) = \infty \quad \text{(infinite-dimensional range)}$$

This is the operator-algebraic statement: the composition P_Π T P_Π projects onto an infinite-dimensional subspace iff there are infinitely many primes p with p+2 also prime.

5. The Angular Gap $\delta\theta_p \to 0$

The geometric picture: prime nodes on $■_L$ cluster near the cusp π. Twin primes are pairs that cluster together.

Theorem 5.1 (Angular gap formula).

For a twin prime pair (p, p+2):

$$|\theta_{p+2} - \theta_p| = 2|\arctan(2 \log p) - \arctan(2 \log(p+2))| \approx 4/(p \log p) + O(1/(p^2 \log p))$$

As $p \to \infty$: the angular gap $|\theta_{p+2} - \theta_p| \to 0$ at rate 4/(p log p). Twin primes approach angular coincidence — they become 'nearly the same node' on $■_L$.

Geometric interpretation: if we zoom into the cusp $w=-1$, twin prime pairs look like nearly-coincident points on the boundary. The question 'are there infinitely many twin primes?' becomes: 'do these nearly-coincident prime pairs continue appearing as we zoom in infinitely?'

6. Zhang-Maynard-Tao in Hologram Language

The landmark results of Zhang (2013) and Maynard-Tao (2014) have a beautiful interpretation on $■_L$.

Theorem 6.1 (Zhang-Maynard in A_L).

There exist infinitely many pairs of consecutive prime nodes $(w_{p_n}, w_{p_{n+1}})$ on $■_L$ with angular gap:

$$|\theta_{p_{n+1}} - \theta_{p_n}| \leq C$$

for a fixed constant $C > 0$. This is the Maynard-Tao theorem (gap ≤ 246) translated to angular gap on $■_L$. The angular gap C corresponds to an integer gap of at most 246 via the formula $\delta\theta \approx 4/(p \cdot 246)$.

Proof sketch.
Maynard-Tao use the GPY sieve (Goldston-Pintz-Y■ld■r■m) to show that for k-tuples of linear forms, at least two are simultaneously prime infinitely often. In A_L: this is the statement that the k-fold prime projection $P_\Pi^{\otimes k}$ has rank ≥ 2 on infinitely many admissible k-tuples. The angular gap bound follows from the admissibility condition on the tuple translated to the $■_L$ metric. ■

The gap from 246 to 2: Maynard-Tao prove that among k ≥ 50 consecutive admissible integers in a window of size 246, at least two are prime infinitely often. To get the gap = 2 specific pair (p, p+2), we need to show the two primes can be chosen to be exactly 2 apart — which requires showing (n, n+2) is admissible infinitely often. This is conjectured but not proved.

7. Brun's Constant as a Trace in A_L

Brun's theorem has a beautiful interpretation as a convergent trace in A_L.

Theorem 7.1 (Brun's Constant as Trace).

The Brun's constant B is:

$$B = \sum_{(p,p+2)\text{ twin primes}} (1/p + 1/(p+2)) = \tau(P_{\Pi 2} + T\, P_{\Pi 2}\, T^*)$$

where $P_{\Pi 2} = \sum_{p:\, p \text{ and } p+2 \text{ both prime}} e_p \otimes e_p^*$ *is the twin prime projection. Brun's theorem:* $\tau(P_{\Pi 2}) = \sum_{p \text{ twin prime}} 1/p = B/2 < \infty$.

In A_L: $B < \infty$ means $P_{\Pi 2}$ is a **trace-class operator**. But trace-class does not mean finite rank. An infinite-dimensional operator can be trace-class. Brun's theorem proves trace-class but not finite rank. Twin prime conjecture asks for infinite rank.

This is the precise distinction: $B < \infty$ (Brun, proved) says $P_{\Pi 2}$ is trace-class. dim range($P_{\Pi 2}$) = ∞ (open) says $P_{\Pi 2}$ has infinite rank. In general: trace-class ⇔ finite rank? No. But for projections: trace-class ⇔ finite-dimensional range. So if $P_{\Pi 2}$ is a projection AND trace-class, then... it has finite rank? This would mean finitely many twin primes!

Careful Remark 7.2.

$P_{\Pi 2}$ as defined above is NOT a bounded projection in the operator norm — it is only trace-class in the τ-sense. The correct statement: $P_{\Pi 2}$ is trace-class in the A_L-trace (meaning $\sum 1/p < \infty$ over twin primes), but this does NOT imply finite rank. This is the subtle point that Brun's theorem does not resolve the conjecture.

8. The Sieve Method as Spectral Truncation

The classical sieve methods translate naturally into spectral truncation in A_L.

Definition 8.1 (Spectral Sieve).

*The **spectral sieve** at level D is the truncated prime projection:*

$$P_\Pi^{(D)} = \sum_{p \text{ prime},\, p \le D} e_p \otimes e_p^*$$

As $D \to \infty$: $P_\Pi^{(D)} \to P_\Pi$. The Brun sieve uses $P_\Pi^{(D)}$ with $D = x^{1/2}$ to estimate #{twin primes $\le x$}.

Theorem 8.2 (Sieve estimate in A_L).

$$\#\{\text{twin prime pairs } (p, p+2) \le x\} = \tau(P_{\Pi 2}^{(x)}) \cdot x \approx 2C_2\, x/(\log x)^2$$

where $C_2 \approx 0.6601618...$ is the twin prime constant. This is the Hardy-Littlewood conjecture (proved asymptotically under GRH, proved as an upper bound by Brun-Titchmarsh). The formula gives infinitely many twin primes IF the asymptotic is exact — but the asymptotic is itself conjectural.

9. The Remaining Gap

We are honest about what is proved and what remains.

Theorem 9.1 (Status of Twin Primes in A_L).

The following are proved in A_L:

(i) Twin primes ⇔ dim range($P_\Pi\, T\, P_\Pi$) = ∞. ✓

(ii) Angular gap $\delta\theta_p \to 0$ as $p \to \infty$ (all pairs become close). ✓

(iii) Brun: $\tau(P_{\Pi 2}) = B/2 < \infty$ (twin primes are sparse). ✓

(iv) Zhang-Maynard-Tao: bounded angular gap $\le C$ infinitely often. ✓

(v) Hardy-Littlewood singular series: $C_2 > 0$ (no arithmetic obstruction). ✓

The following remains open:

(vi) dim range(P_Π T P_Π) = ∞ with T = shift-by-exactly-2. ✗ **OPEN**

The gap between 'bounded gaps infinitely often' and 'gap = 2 infinitely often': Maynard-Tao get gap ≤ 246. To get gap = 2, we need to show the specific pair (p, p+2) is admissible in the Maynard-Tao sense. This requires showing that the set {0, 2} is admissible — i.e., for every prime q, not every residue class mod q is covered by {0, 2}. For q = 2: {0, 2} covers 0 and 0 mod 2 — both even. So {0, 2} is NOT admissible! This is why (p, p+2) cannot be directly handled by Maynard-Tao without additional input.

Remark 9.2 (The parity obstruction).

The set {0, 2} is not admissible because for p > 2, p must be odd, and p+2 must also be odd — but (odd) + 2 = odd only if... actually p odd and p+2 odd means p ≡ 1 (mod 2) so p+2 ≡ 1 (mod 2) — this is fine. The issue is mod 3: if p ≡ 1 (mod 3) then p+2 ≡ 0 (mod 3), not prime. So twin primes must have p ≡ −1 (mod 3), i.e., p ≡ 2 (mod 3). This is the 'parity obstruction' — sieve methods have trouble with it. In A_L: this is the interaction between P_Π, T, and the Folding Tower at p=3.

Twin Primes in A_L — Summary

Twin primes are pairs (w_p, w_{p+2}) on ■$_L$ that are simultaneously prime nodes with angular gap → 0. Brun: the total weight B ≈ 1.902 is finite — twin primes are trace-class. Zhang-Maynard: gaps ≤ 246 infinitely often — consecutive prime nodes come close infinitely often. The conjecture: they come within angular gap δθ = 4/(p log p) of each other infinitely often as p and p+2 prime pairs. The obstacle: the parity-mod-3 constraint filters which prime pairs (p, p+2) are admissible. A_L sees this as the interaction between the Twin Operator T and the Folding Tower at prime 3.

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